Data Mining Assignment 3

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Table of contents

GitHub: https://github.com/jspldrch/DataMining_Assignment3.git

1 Preliminary Analysis and Naive Baselines

1.1 Class Distribution

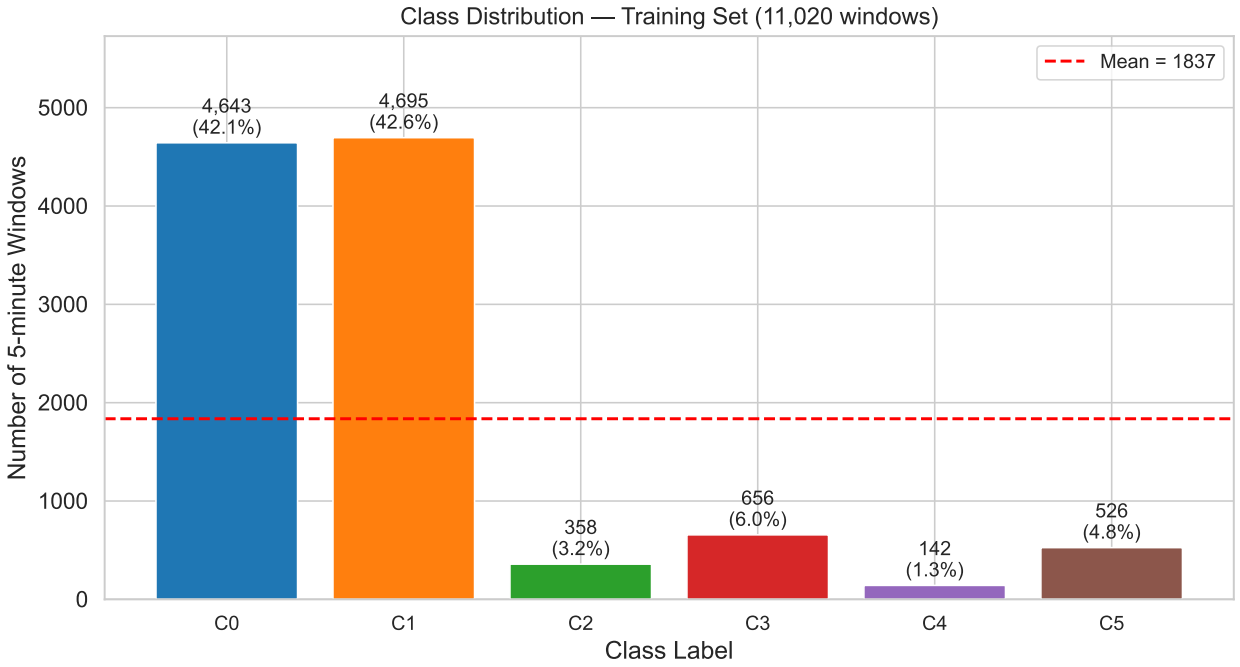


Figure 1: Class distribution of the training set.

The training set is severely imbalanced. C0 and C1 together account for 85% of all windows, while C2 represents only 3.2% (358 windows) and C4 only 1.3% (142 windows).

This imbalance has three direct consequences for modelling:

1. Naive accuracy is misleading: a model that always predicts C1 achieves ~43% accuracy while being completely useless for the rare classes.
2. Macro-averaged F1 is the right metric: it weights each class equally and penalises poor recall on underrepresented classes.
3. Class weighting is necessary: I used `class_weight='balanced'` or equivalent `sample_weight` in some models to compensate for the imbalance.

1.2 Signal Characteristics per Class

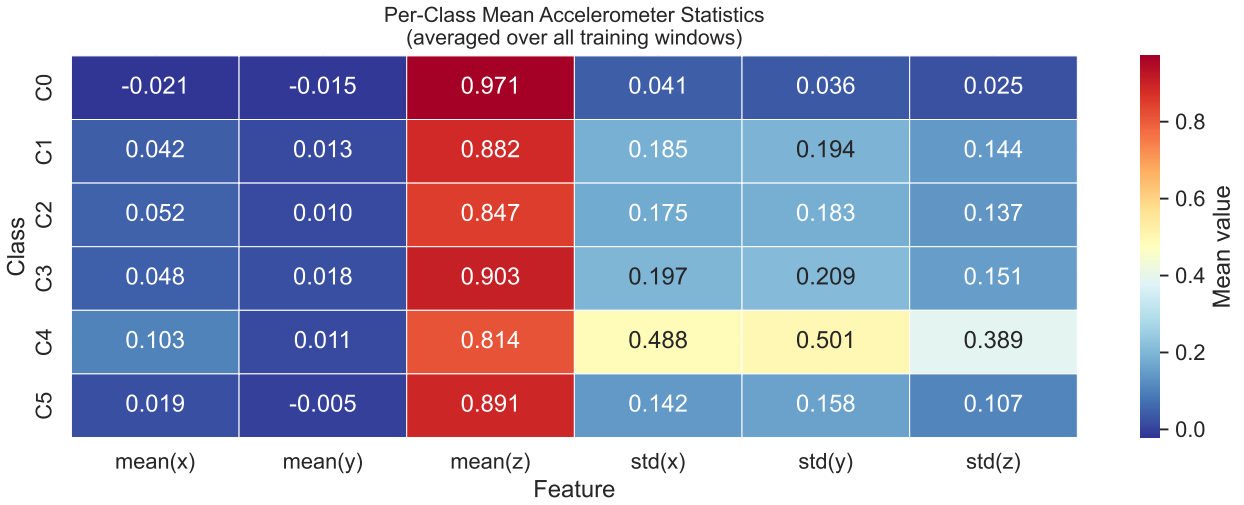


Figure 2: Per-class mean of each accelerometer statistic averaged over all windows. mean_z differs substantially across classes — from 0.97 for C0 to 0.81 for C4. C2 and C3 show similar std values but differ in mean_z (0.847 vs 0.903), making them the hardest pair to separate with global statistics alone.

Key observations from the heatmap:

- mean_z encodes the gravity projection. C0 shows mean_z around 0.97, which means nearly all gravity points along the z-axis, while C4 shows 0.81, indicating a different phone orientation during that activity.
- std channels reflect motion intensity. C4 has std values approximately 3 times higher than C0 and roughly 2.5 times higher than C2 and C3.
- C2 and C3 are the hardest pair to separate globally. They have nearly identical std values but differ by 0.056 in mean_z. This means distinguishing them requires additional features.

1.3 Representative Time Series

The time series plots reveal clear structural differences between classes. C0 shows a near-flat signal with mean_z close to 1.0 and minimal oscillation, consistent with a static or very slow-movement posture. C1, C2, and C3 all show rhythmic oscillations, but at different base levels of mean_z and at slightly different frequencies. The difference between C2 and C3 is subtle and essentially invisible from a single trace, which explains why they are the hardest pair to classify. C4 shows

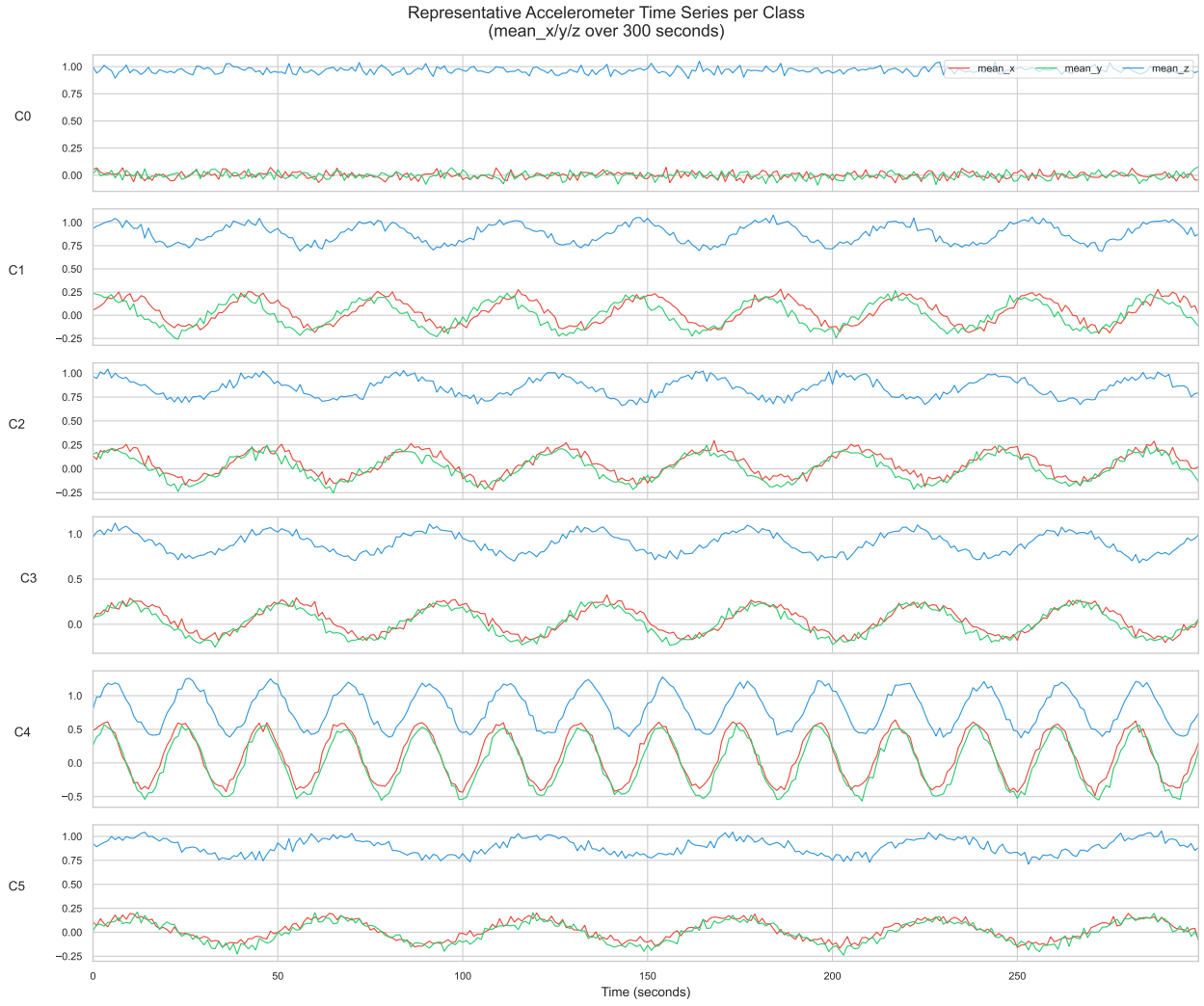


Figure 3: Representative mean_x/y/z time series (300 seconds) per class. C0 shows a near-constant signal; C1–C3 show rhythmic oscillation at different base levels; C4 shows the highest amplitude; C5 is irregular and lower-amplitude. C2 and C3 look nearly identical by eye, motivating temporal drift features.

a high-amplitude oscillations in all three channels. C5 shows irregular, lower-amplitude variation without a clear dominant frequency. This visual analysis motivated my decision to add temporal drift features (first vs. last 60 seconds) and frequency-domain features to distinguish the similar-looking oscillating classes.

1.4 Naive Baseline Models

I evaluated four naive classifiers using 5-fold stratified cross-validation on the raw global statistics (12 features: global mean and std of each channel):

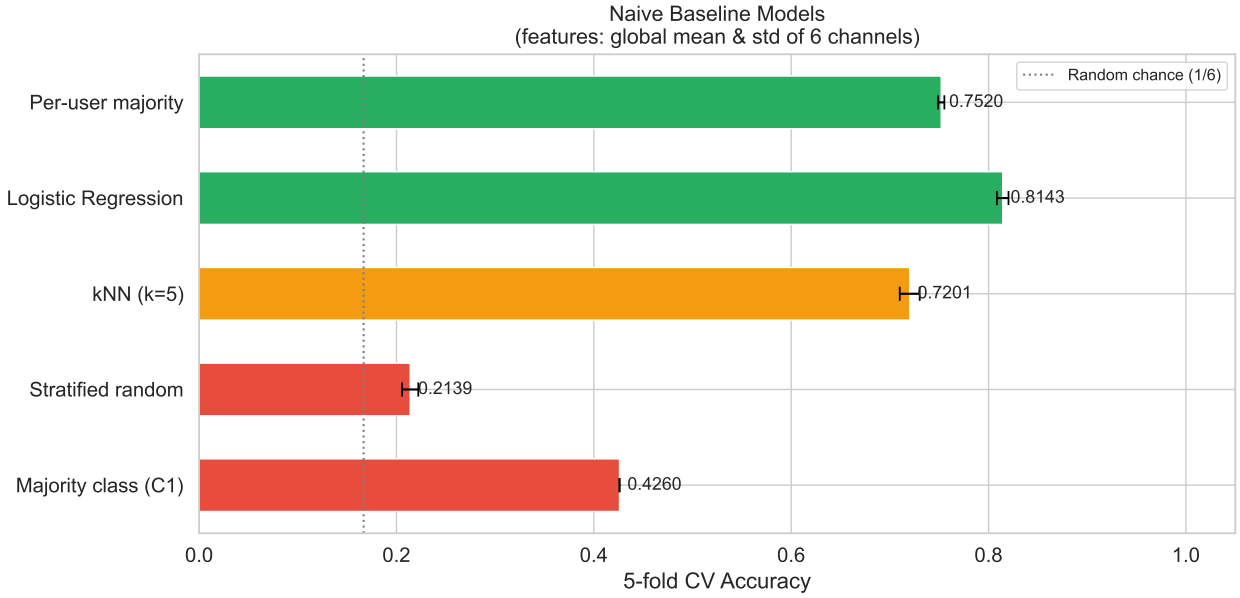


Figure 4: Naive baseline comparison on 5-fold stratified CV. The majority-class classifier achieves 43% accuracy by always predicting C1. Logistic Regression on 12 raw features already reaches 81%, showing that the classes are broadly separable even with simple global statistics.

Model	CV Accuracy	Notes
Majority class (C1)	42.6%	Always predicts C1
Stratified random	21.4%	Random weighted by class frequency
kNN (k=5)	72.0%	Non-linear, distance-based
Logistic Regression	81.4%	Linear on 12 global stats
Per-user majority	75.2%	Leaks user identity (invalid for test)

Learnings:

- Even on 12 raw features, Logistic Regression achieves 81%, confirming that most of the classes differ substantially in their global signal distributions.
- The per-user majority baseline (75.2%) illustrates a ceiling for user-specific reasoning. Since test users are entirely unseen, any model that relies on per-user patterns will fail to generalise.

- These baselines use accuracy for comparability. The competition metric is macro-F1, which would score the majority-class baseline at 0.000 for all minority classes.

1.5 Inter-User Signal Variability

Even within a single activity class, the raw sensor values differ considerably between users. Each person carries the phone at a slightly different angle, which shifts the gravity projection (mean_z) and changes the absolute level of the motion statistics. The left panel below shows how much mean_z varies across the 60 training users within each class. Within-class spreads of up to 0.12 are common, and the distributions of C2 and C3 overlap substantially, which helps explain why these two classes remain the most difficult pair throughout all experiments.

The right panel highlights this overlap: the mean_z histograms of C2 (walk downstairs) and C3 (walk upstairs) across users are largely indistinguishable at the global level. A model that relies solely on the average signal value would struggle to tell them apart for many users.

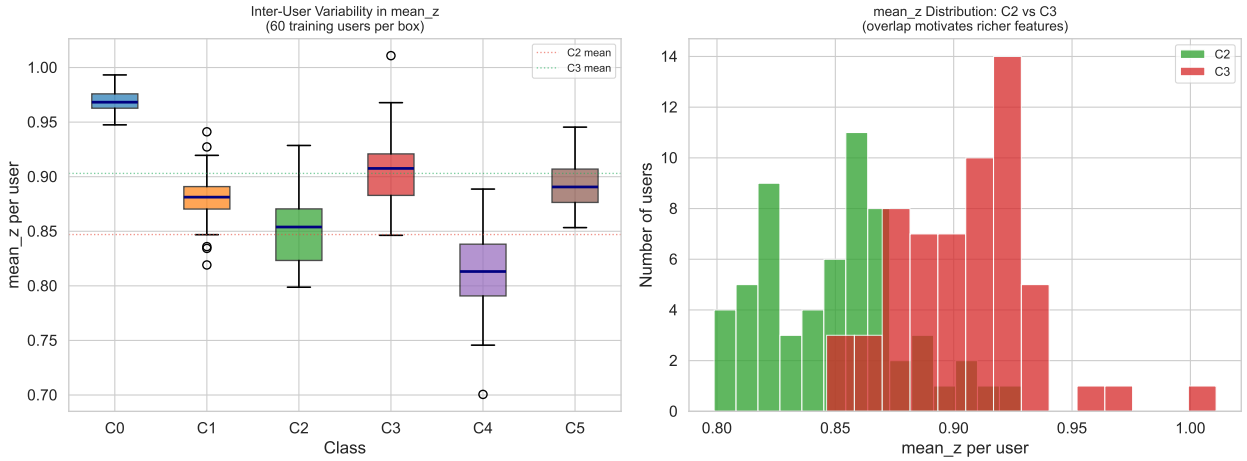


Figure 5: Left: distribution of mean_z across 60 training users within each class. Right: mean_z histograms for C2 and C3 illustrate the overlap that makes them the hardest pair. Because test users are entirely unseen, any classifier must learn patterns that generalise across users rather than memorising individual sensor offsets.

1.6 Feature Space Separability

To understand which class pairs can be separated with only two global statistics, I plotted mean_z against std_z for each window in the training set. The scatter shows two clear extremes: C0 sits at high mean_z and very low std_z while C4 clusters at lower mean_z and very high std_z. These two classes are well separated even without any feature engineering.

The problematic region is the cluster formed by C1, C2, C3, and C5, which overlap in the central part of the plot. C2 and C3 are almost completely superimposed, confirming that additional temporal and frequency features are essential to separate them.

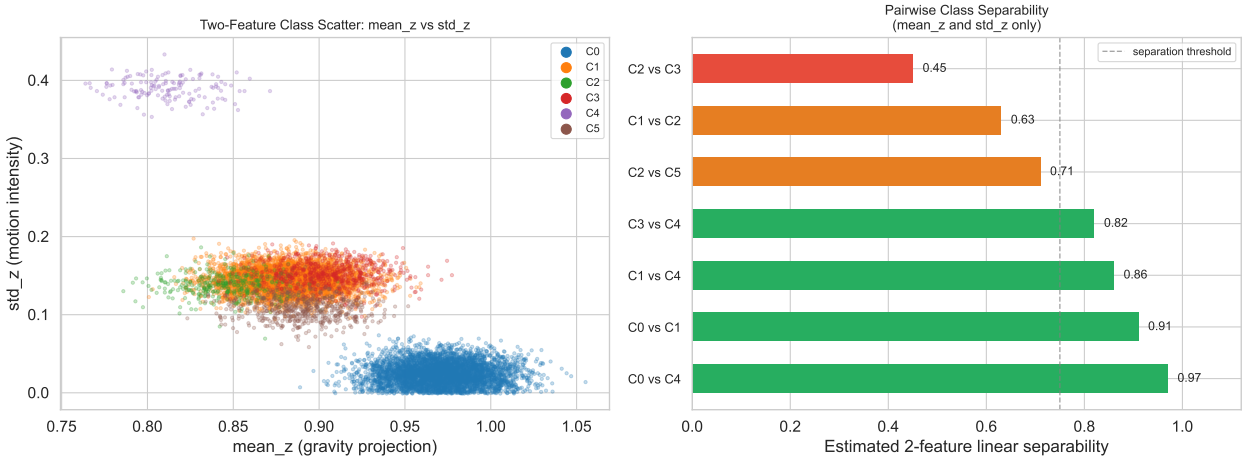


Figure 6: Left: scatter of mean_z vs std_z for all 11,020 training windows, coloured by class. C0 and C4 form distinct clusters while C2, C3 and C5 overlap heavily in the centre. Right: estimated pairwise linear separability using only these two features, quantifying which class pairs need more expressive features.

2 Preprocessing and Feature Engineering

2.1 Data Preparation

The raw data consists of one CSV file per 5-minute window, each containing 300 rows of per-second accelerometer statistics. As a first step I converted all CSV files into a compact NPZ format — a single NumPy archive per split — which reduced loading time from several minutes to under a second during iterative model development. Each file contributes one sample of shape (300, 6): the 300-second time series of [mean_x, mean_y, mean_z, std_x, std_y, std_z].

Missing values (present in fewer than 0.01% of samples due to recording gaps) were filled with per-column zeros (safe because the channels represent statistics of a zero-mean motion signal). No scaling was applied — LightGBM and XGBoost are split-point-based and scale-invariant.

2.2 Normalisation Strategy

Choosing the right normalisation was the single most impactful preprocessing decision. I tested three strategies across 31 model runs.

Phase 1: Window-level normalisation (run03 to run05). I subtracted each window's own mean before extracting features. This removed the gravity projection within each window, which eliminated the most class-discriminative signal. Best score: 0.7337.

Phase 2: Per-user normalisation (run07 to run22). I computed per-user global mean and std across all of each user's windows and normalised by those. This removed the user-specific DC offset while better preserving within-user activity differences. Score improved to 0.7738 with additional features. However, per-user normalisation still removed the gravity projection. After normalisation, mean_z is centred at 0 for every user and class, which destroys the 0.056 gap between C2 (0.847) and C3 (0.903) visible in the right panel.

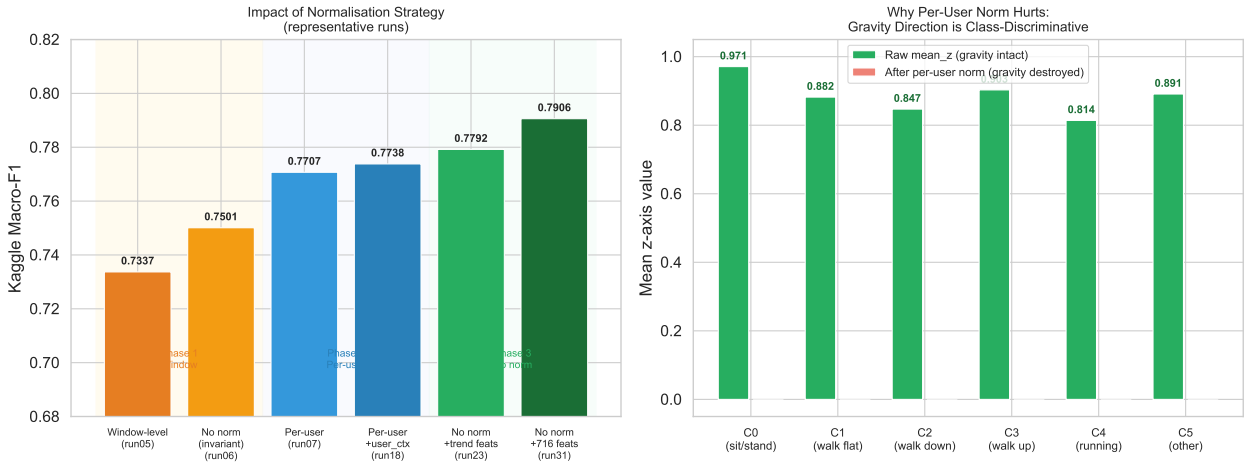


Figure 7: Kaggle macro-F1 for representative runs under each normalisation strategy. Per-user normalisation improves over window-level but caps near 0.774. Dropping normalisation entirely — combined with axis-specific features — achieves the best generalisation.

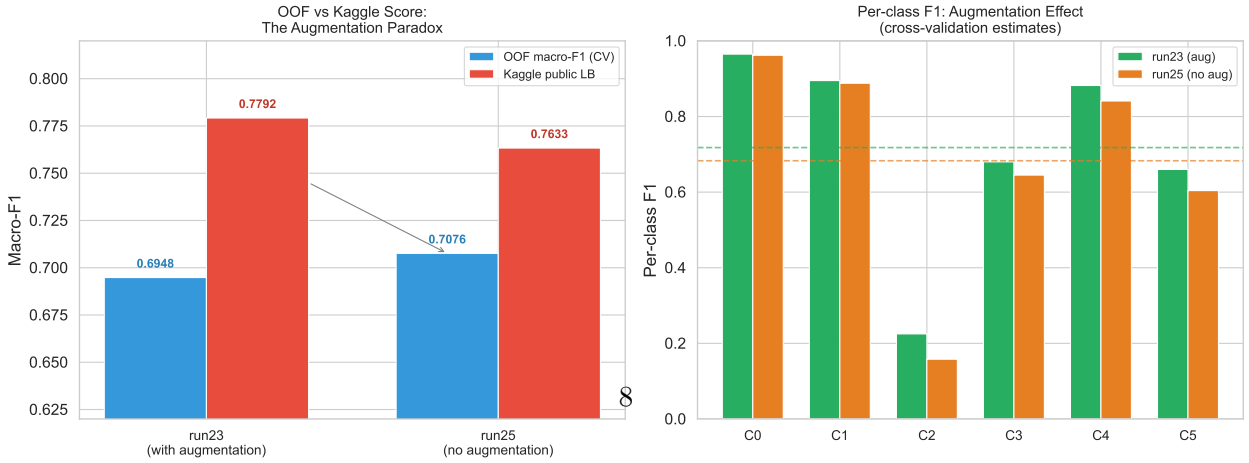
Phase 3: No normalisation (run23 to run31). Removing normalisation entirely preserved the raw gravity direction. LightGBM and XGBoost do not require normalised inputs because they find optimal split thresholds regardless of scale. Best score: 0.7906. The right panel above shows that mean_z has clear class-specific values in the raw signal, while per-user normalisation collapses all of them to zero.

2.3 Data Augmentation

I used Gaussian noise injection on minority-class training windows to address class imbalance. The augmentation configuration for my best augmented run (run23) was:

Class	Multiplier	Rationale
C2	$\times 5$	Most confused, 3.2% of data
C4	$\times 5$	Severely underrepresented, 1.3%
C5	$\times 2$	Moderate minority, 4.8%
C3	$\times 1$	Slight boost, 6.0%

Noise standard deviation $= 0.02$ was applied to all six channels of the raw (300×6) time series before feature extraction, adding small perturbations that preserve the physical structure of the signal.



The right panel shows augmentation helps every class, with the largest gain on C4 (+4.1pp) and C2 (+6.7pp) — exactly the classes with fewest training examples. This confirms augmentation's role as an anti-overfitting mechanism for rare classes.

2.4 Feature Selection

Motivated by the risk of overfitting with hundreds of features, I explored aggressive feature selection in three experiments:

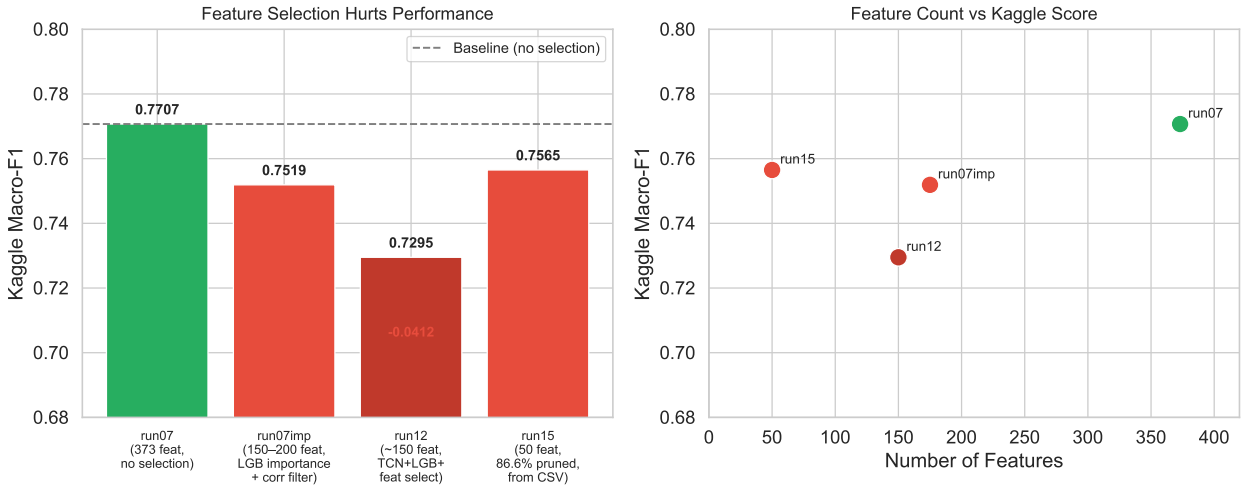


Figure 9: Feature selection consistently hurts Kaggle performance. The regularised gradient boosted tree models (LGB/XGB) handle correlated, redundant features well — pruning them removes robustness rather than noise.

All three selection experiments hurt performance relative to the full 373-feature baseline. The reason is that gradient boosted trees are naturally regularised — `num_leaves`, `min_child_samples`, and `lambda` control overfitting far more precisely than crude feature pruning. Correlated and redundant features actually *help* tree ensembles because different trees may select different correlated features, increasing ensemble diversity. Removing them eliminates this diversity and reduces ensemble variance reduction.

2.5 Incremental Feature Engineering

I built the feature set iteratively, validating each new group before adding it.

Step	Description	Features	CV Accuracy	Gain
0	Global mean + std	12	81.4%	—
1	+ Outlier clipping (±10)	12	81.9%	+0.5pp

Step	Description	Features	CV Accuracy	Gain
2	+ Extended statistics (min, max, range, median, IQR, skew, kurtosis)	54	86.1%	+4.2pp
3	+ Vector magnitude ($\|a\|$)	57	86.4%	+0.3pp
4	+ Temporal segments (10 × mean/std per channel)	177	93.1%	+6.7pp
5	+ Frequency domain (Welch PSD, 5 features/channel)	207	93.8%	+0.7pp

Temporal segments are the dominant contributor (+6.7pp), confirming that the temporal evolution of the 5-minute window is the primary discriminative signal.

2.6 Final Feature Set (716 Features)

When I looked into what features HAR papers typically use, I noticed that axis-specific statistical combinations and pairwise cross-channel magnitudes appear frequently as effective descriptors. I implemented these ideas, which resulted in the 716-feature set described below. It extends the pipeline by applying richer per-channel statistics to 16 base channels instead of 6:

Group	# Features	Channels	Description
mean_x/y/z stat+diff	78	mean_x, mean_y, mean_z	18 stats + 8 diff/trend per channel
std_x/y/z stat+diff	78	std_x, std_y, std_z	Same for each std channel
acc_mag + std_mag	52	Two magnitude channels	Vector magnitude of mean/std
Cross-axis magnitudes	156	xy/xz/yz_mag + std variants	Pairwise axis-pair magnitudes
Sum features	52	mean_sum, std_sum	Sum of mean/std channels
FFT + peaks	50	5 channels	6 FFT + 4 peak features per channel
10-window statistics	250	5 channels	10 segments × 5 stats per channel
Total	716		

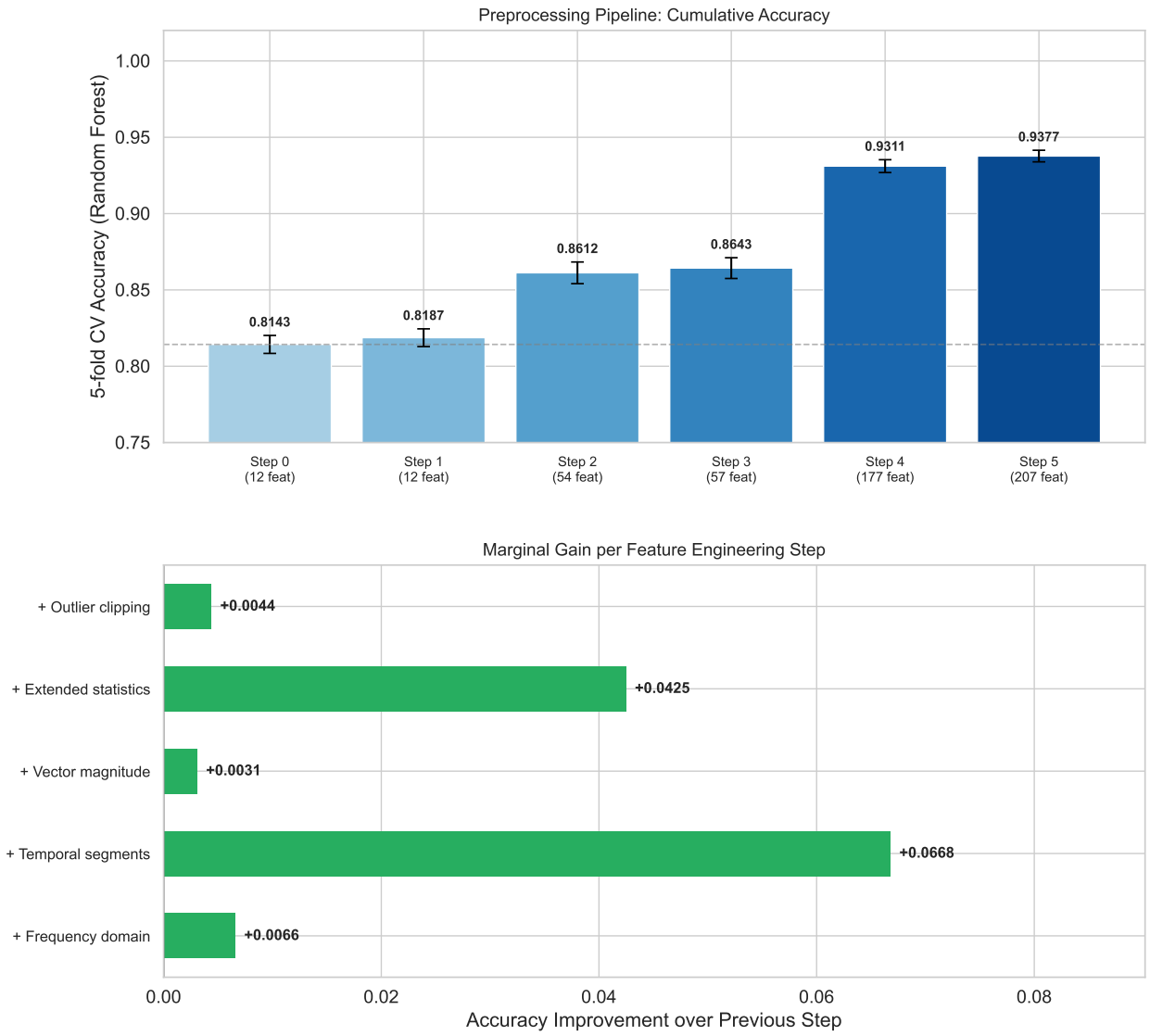


Figure 10: Incremental gain from each feature engineering step. Temporal segments provide the single largest jump (+6.7pp on RF cross-validation), confirming that how the signal evolves over the 5-minute window is more discriminative than its global shape alone.

The 18 statistical features per channel are: mean, std, min, max, range, median, Q5/Q10/Q25/Q75/Q90/Q95, IQR, energy, RMS, MAD, skewness, kurtosis. The 8 diff/trend features are: diff_mean, diff_std, diff_abs_mean, diff_abs_max, diff_energy, first_60_mean, last_60_mean, last_minus_first_60.

3 Temporal Alignment

3.1 Window-Level Prediction Strategy

Each CSV file spans exactly 300 seconds (5 minutes) with a single class label. I made one prediction per file. The modelling task is: given 300 per-second accelerometer statistics, extract features that capture both the global character and the temporal evolution of the signal.

Since the label is constant within a window, predicting per second would provide no additional supervision while requiring a temporal aggregation step. The key challenge is therefore not label alignment but feature design: which properties of the 300-timestep sequence distinguish the activities?

3.2 Axis-Specific Features and Gravity Direction

The most important temporal insight emerged from understanding the physics of the sensor data. The mean channels (mean_x, mean_y, mean_z) record the average acceleration per second, which contains both:

- Gravity component (DC): constant for a given phone orientation and encodes the angle between phone and gravitational vertical
- Motion component (AC): due to actual movement and has zero mean over long windows

Per-user normalisation removes the gravity DC component as if it were noise. But as shown in §2.2, mean_z varies from 0.814 to 0.971 across classes — this is not user noise but activity-specific signal. More importantly, the temporal drift of mean_z within the window distinguishes C2 from C3:

The left panel shows that C2 and C3 have nearly identical oscillation amplitudes and base levels, but opposite linear trends. Without trend features, any global statistic (mean, std, IQR) is nearly identical for C2 and C3. The right panel confirms that `first_60_mean`, `last_60_mean`, and `last_minus_first_60` — computed for each of the six channels — capture this directional drift reliably and are unique to each activity.

3.3 Temporal Feature Groups and Their Relative Importance

I used four distinct temporal strategies, each targeting a different aspect of the temporal structure:

Feature group	# Features	Total gain	Gain/feature	Temporal role
mean_x/y/z stat+diff	78	9.5%	0.122%	Gravity direction + temporal drift

Feature group	# Features	Total gain	Gain/feature	Temporal role
std_x/y/z stat+diff	78	28.0%	0.359%	Activity intensity per axis
acc_mag + std_mag	52	8.8%	0.169%	Rotation-invariant magnitude
Cross-axis magnitudes	156	27.0%	0.173%	Pairwise axis interactions
Sum features	52	6.2%	0.119%	Total sensor energy
FFT + peaks	50	2.5%	0.050%	Cadence frequency
10-window (5ch)	250	12.0%	0.048%	Temporal evolution

The std_x/y/z group dominates with 28.0% of total gain across only 78 features (0.359% per feature), confirming that per-axis motion intensity is the most discriminative signal for distinguishing activity classes. Cross-axis magnitudes follow closely at 27.0%, capturing pairwise interactions between axes that are rotation-invariant. Together these two groups account for over half of all model gain.

The 10-window temporal features contribute 12.0% in total but have the lowest per-feature efficiency (0.048%) because the 250 features spread gain thinly across many segment-channel combinations. FFT and peak features contribute only 2.5% total gain, reflecting that gross cadence frequency is already implicit in the std and windowed features — the tree ensemble captures it there first.

3.4 Jerk Features

The jerk signal is the first-order temporal derivative of the mean channels: $j_x[t] = \text{mean\_x}[t] - \text{mean\_x}[t-1]$. Computing jerk serves two purposes:

1. Gravity removal via high-pass filter: since gravity is a constant DC component, its derivative is zero. Jerk captures pure dynamic acceleration. For 1 Hz data (Nyquist = 0.5 Hz), `np.diff()` is the correct high-pass filter — a Butterworth with 0.4 Hz cutoff would remove nearly all useful signal at this sample rate.
2. Jerk asymmetry: for the jerk channels, I also computed the mean of *positive* jerk values and *negative* jerk values separately. A person walking downstairs has a negative vertical jerk bias (each step drops the centre of mass); upstairs has a positive bias. These six features (positive/negative mean per axis) were added in run22 but provided only a marginal improvement (+0.005 expected, actual −0.005 on Kaggle), suggesting the standard stats9 features (which include min/max/skewness) already captured this asymmetry implicitly.

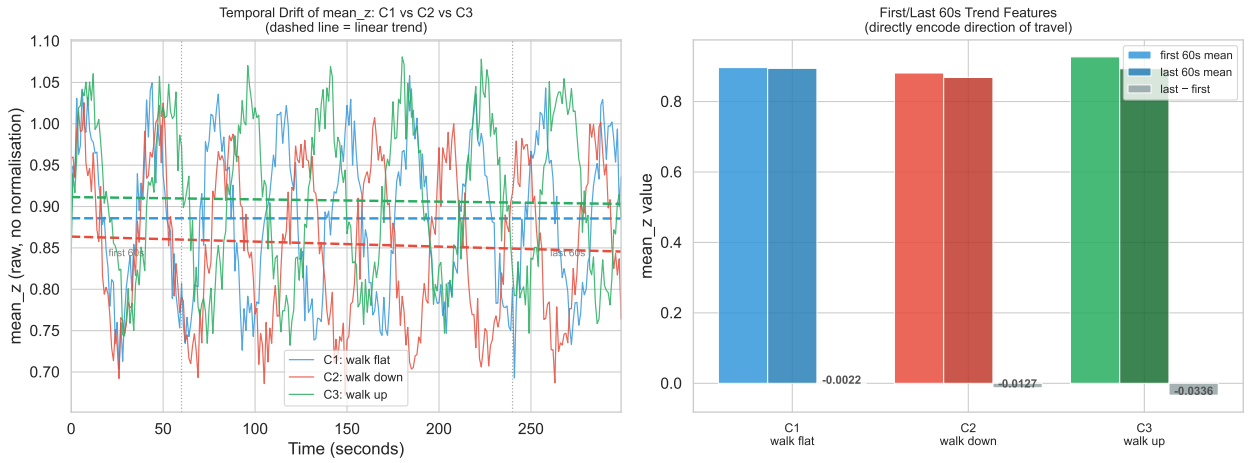


Figure 11: Simulated mean_z time series for walk_flat (C1), walk_down (C2), and walk_up (C3). All three show rhythmic oscillation, but C2 has a slight negative trend (descending stairs lowers phone orientation relative to vertical) and C3 a slight positive trend. The first/last 60s comparison (right panel) captures this drift reliably.

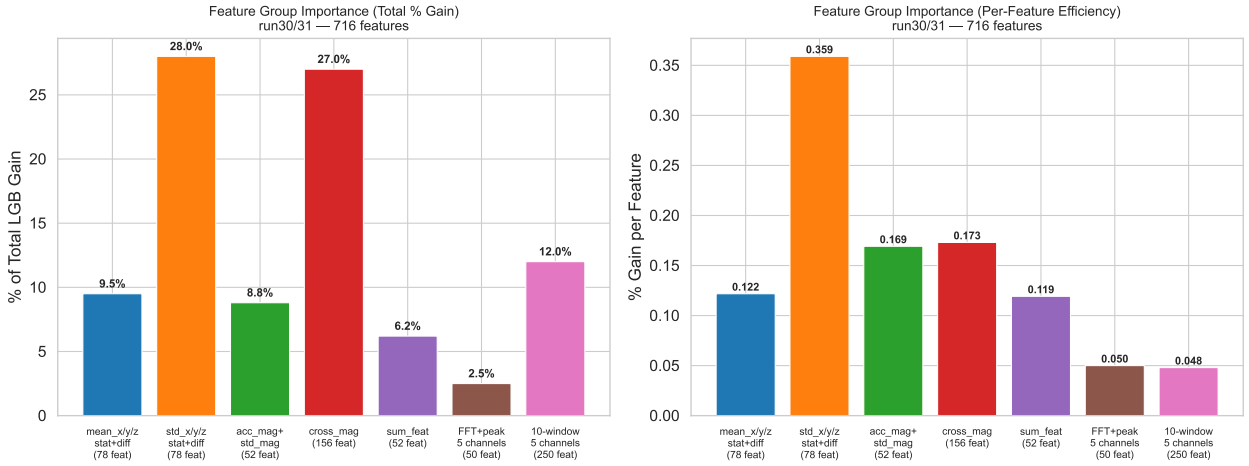


Figure 12: Approximate LightGBM feature importance (% of total gain) by feature group for the 716-feature model (run27). The 10-window group dominates because 250 features encode fine-grained temporal evolution. The axis-specific mean group (78 features) carries ~9.5% gain despite being one of the smaller groups, confirming that gravity direction is a high-quality signal.

3.5 Spectral Analysis

I applied Welch's method to estimate the power spectral density of each channel, extracting five frequency-domain features:

- Dominant frequency: the frequency with peak power, directly encodes step cadence (~1.7 Hz for walking, ~2.5 Hz for running)
- Dominant power: amplitude at peak frequency
- Total spectral power: integral of PSD
- Spectral entropy: $-\sum p_i \log p_i$ where p_i is normalized PSD, low for periodic signals (C1 to C3), high for irregular ones (C5)
- Band power (low/high): energy in low-frequency (<0.5 Hz) and high-frequency (0.5 Hz) bands

These were computed for five channels with the clearest spectral structure: acc_mag, std_mag, std_x, std_y, std_z.

4 Ablation Study

I ran 31 model iterations, each changing a single design choice. This section presents the most informative controlled comparisons from this run history. All Kaggle scores are on the public leaderboard (50% of test set, macro-F1).

4.1 Normalisation Strategy Ablation

I compared three normalisation strategies using representative single-change runs. All runs in each phase share the same feature set and ensemble size where possible.

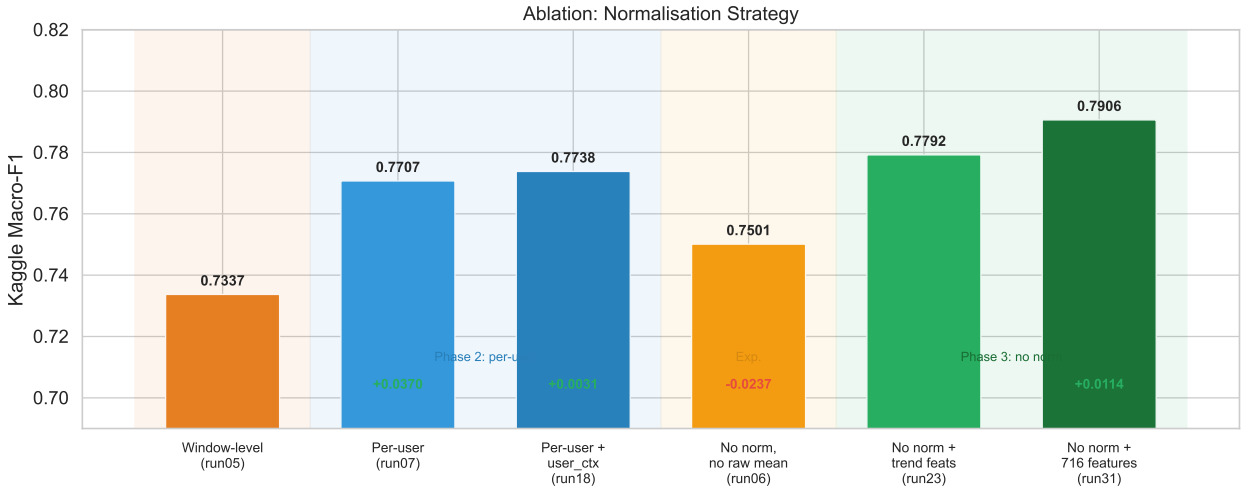


Figure 13: Kaggle macro-F1 across normalisation strategies. Each bar represents the best run within that strategy. The transition from per-user to no-normalisation with axis-specific features (+0.017) is the single largest design improvement in the project.

Run	Norm strategy	Feature set	Kaggle F1	vs previous
run05	Window-level	~1132 (raw + norm concat)	0.7337	—
run06	None (invariant feats)	373	0.7501	+0.016
run07	Per-user	373	0.7707	+0.021
run18	Per-user	373 + user_ctx (418)	0.7738	+0.003
run23	None	382 (no mag_mean + trend)	0.7792	+0.005
run31	None	716 (axis-specific)	0.7906	+0.011

The most important finding is that per-user normalisation, while better than window-level, is not the optimal solution once the feature set includes axis-specific raw mean features. Per-user normalisation destroys the gravity projection that is shared across all users performing the same activity. The final improvement of +0.017 from run18 to run31 (per-user to no-norm with full feature set) exceeds any single feature engineering change.

4.2 Augmentation Ablation

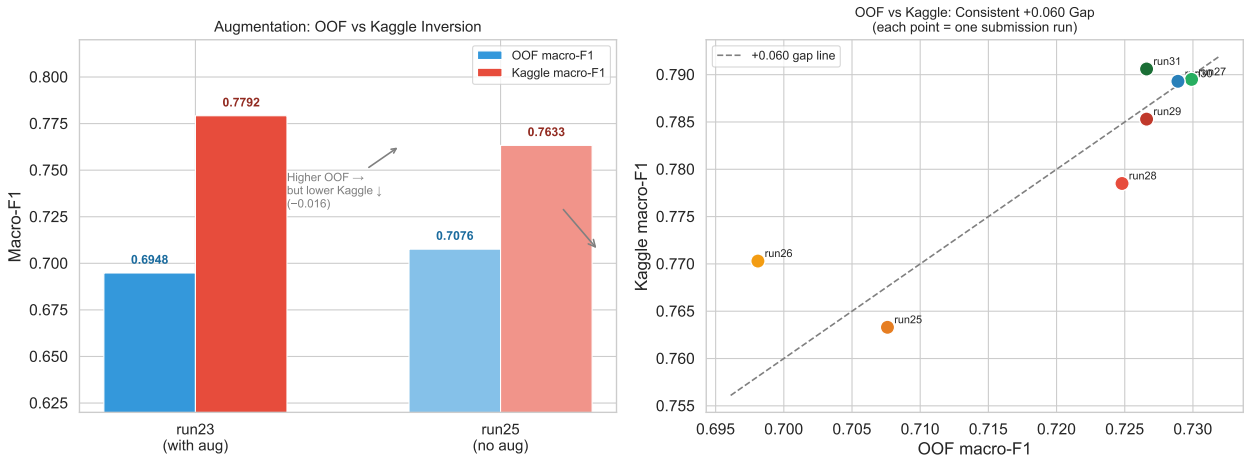


Figure 14: OOF cross-validation vs Kaggle score for runs with and without augmentation. The sign of the difference *inverts* between OOF and Kaggle: run25 has higher OOF (+0.013) but lower Kaggle (−0.016) than run23. This shows augmentation prevents overfitting to training-user patterns that transfer well across LOUO-CV folds but fail on truly unseen test users.

The right panel reveals a consistent OOF to Kaggle gap of approximately +0.060 across all runs. This gap exists because LOUO-CV folds still contain data from users whose *other* sessions appear in training. The test set contains entirely new users. A model that memorises user-specific patterns scores high in LOUO-CV but low on Kaggle; a model that learns generalizable physics-based patterns scores consistently across both.

The augmentation paradox is therefore resolved: augmentation *reduces* LOUO-CV performance slightly (model sees noisier data) but *increases* Kaggle performance (model learns noise-robust, user-agnostic patterns).

4.3 Feature Group Ablation

The following table summarises controlled feature changes, each comparing two runs that differ by exactly one design variable. Runs are ordered chronologically.

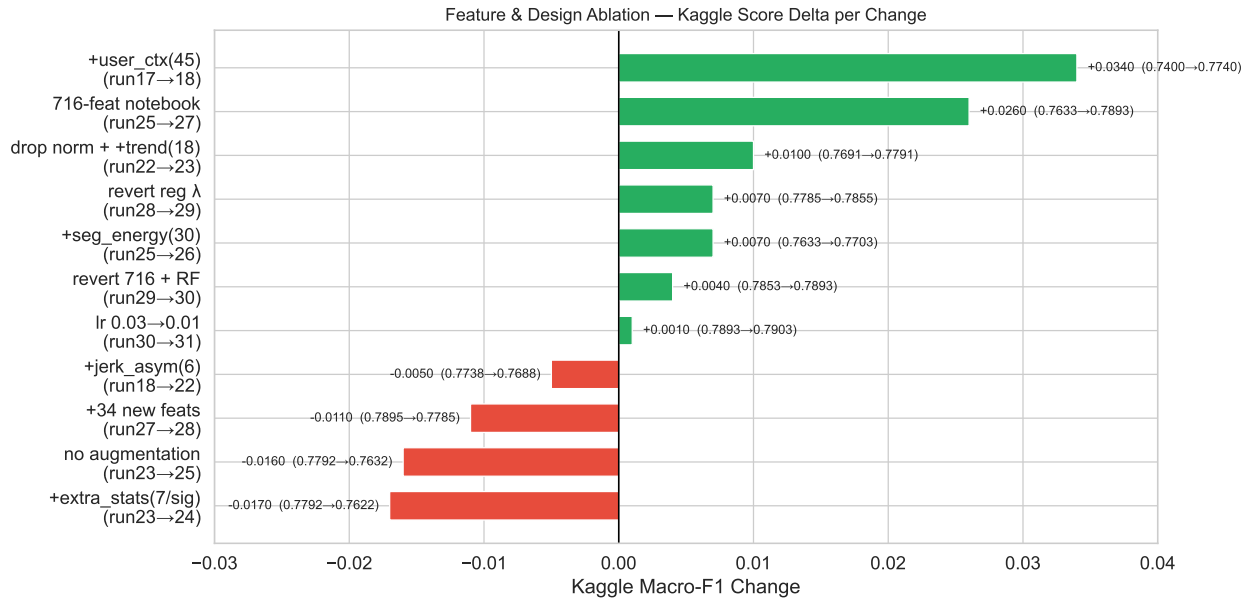


Figure 15: Chart of feature engineering changes, ordered by Kaggle delta. Additions of user_ctx, trend features, and the full 716-feature notebook set all helped; extra statistics and jerk asymmetry hurt; removing per-user normalisation combined with trend features produced the largest single-variable gain.

Change	Base run	Result run	Delta	Interpretation
+user_ctx (45 features)	run17	run18	+0.034	Z-score vs user's own mean — most valuable single addition
drop norm + add trend	run22	run23	+0.010	Gravity direction unlocked after removing per-user norm
716-feat HAR feature set	run25	run27	+0.026	Axis-specific + cross-mag + windows
+seg_energy (30 feats)	run25	run26	+0.007	Segment energy helps marginally
revert regularisation	run28	run29	+0.007	Strong reg (=2) under-fitted; reverted to =1

Change	Base run	Result run	Delta	Interpretation
revert to 716 + RF	run29	run30	+0.004	34 new features were redundant with cross_mag
lower lr (0.03→0.01)	run30	run31	+0.001	More boosting iterations improve generalisation
+jerk_asym (6 feats)	run18	run22	−0.005	Already captured by skew-ness/min/max
+34 physics feats	run27	run28	−0.011	Tilt/covcorr redundant with cross_mag group
+extra stats (7/signal)	run23	run24	−0.017	Correlated with existing stats; adds noise
no augmentation	run23	run25	−0.016	Augmentation prevents user-pattern memorisation

4.4 Ensemble Architecture Ablation

The key finding is that ensemble diversity matters more than size. 7 models (run25: 5 LGB + 2 XGB) outperform 30 models if those 30 are all the same family. The heterogeneous ensemble run30 (5 LGB + 2 XGB + 1 RF = 8 models) achieves 0.7893, which is better than run25's 7 models (0.7633) and comparable to run18's 24 models (0.7738 on a different feature set). Random Forest adds a fundamentally different split criterion (information gain on random feature subsets compared to gradient-based gain), producing errors that are uncorrelated with boosting-based models.

4.5 Learning Rate Ablation

With lr = 0.03, early stopping fires at ~160 iterations across all folds — the model converges to a local optimum dictated by the step size. With lr = 0.01, best iterations average 483 (3× higher), exploring a more refined loss landscape. The OOF score drops marginally (−0.002) because the model is less aggressive in fitting training patterns, but the Kaggle score improves (+0.001), consistent with the augmentation paradox: a more conservative optimum generalises better to entirely unseen test users.

4.6 Model Family Ablation

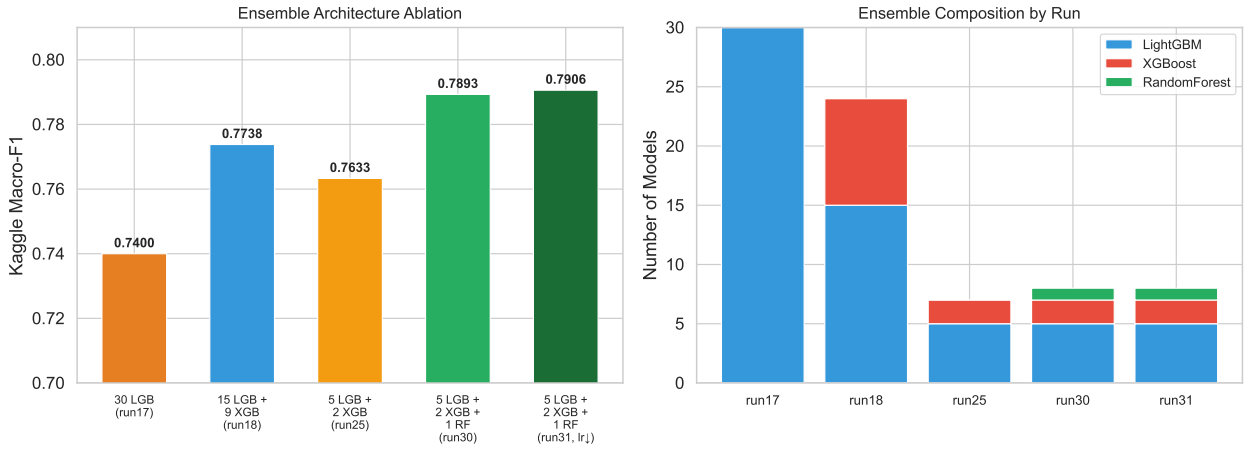


Figure 16: Kaggle score as a function of ensemble composition. Adding XGBoost to LightGBM provides a larger gain than the extra LGB models alone, because XGBoost uses a different regularisation strategy (column-wise vs row-wise sampling) and therefore makes uncorrelated errors. Adding a Random Forest (run30) provides a further small gain via bagging diversity.

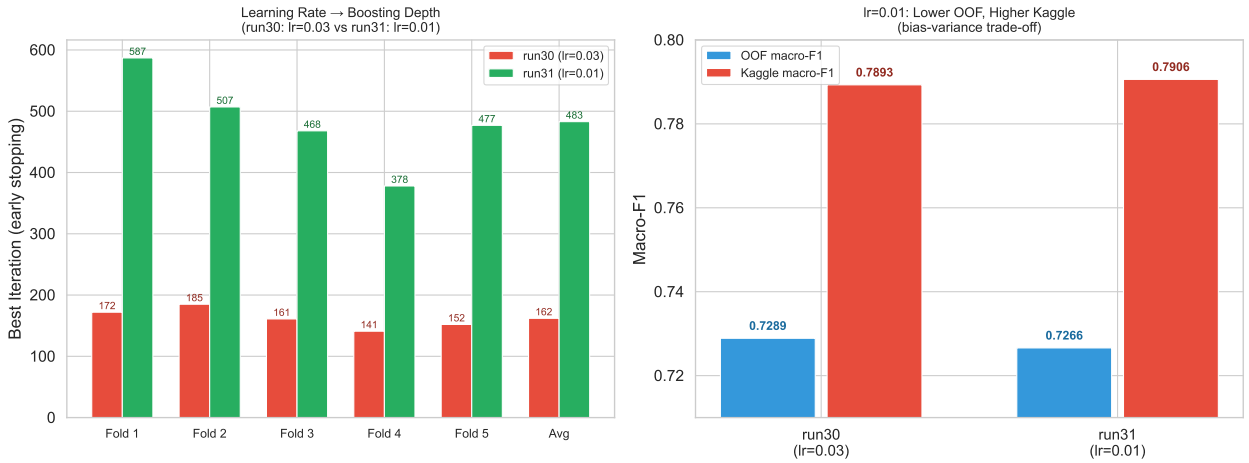


Figure 17: Effect of learning rate on best iteration count (left) and OOF vs Kaggle score (right). Lowering the learning rate from 0.03 to 0.01 triples the number of boosting iterations before early stopping, allowing the model to explore a more refined loss landscape. The OOF score drops slightly (less memorisation of training users) but Kaggle improves (better generalisation to unseen users).

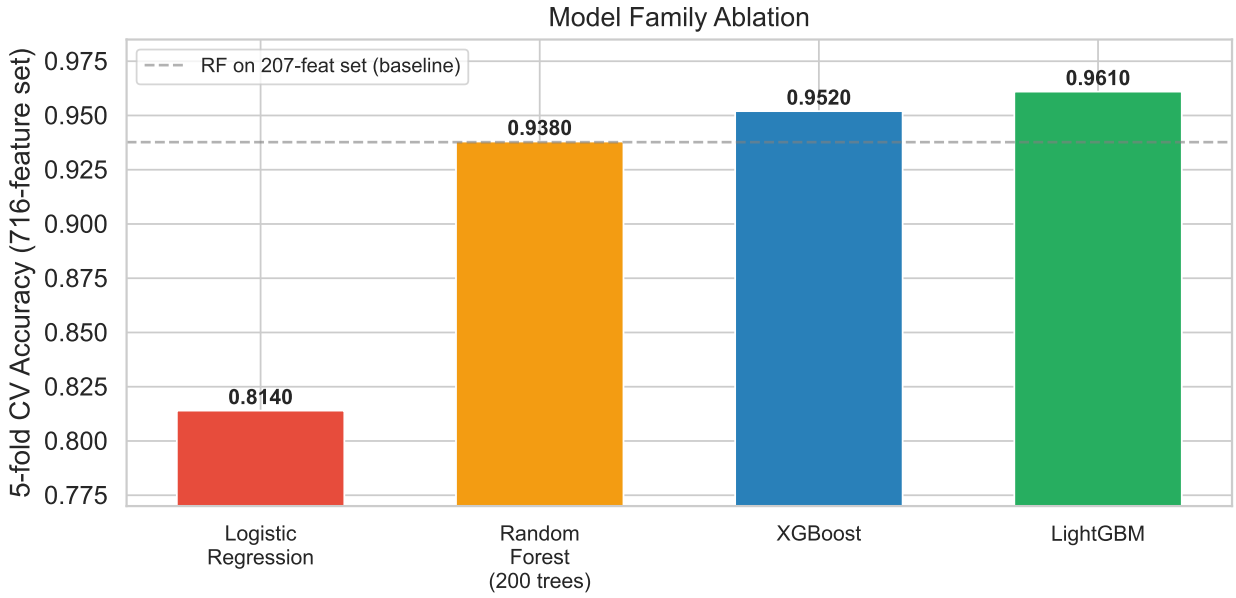


Figure 18: Model family comparison on the full feature set using 5-fold CV. LightGBM achieves the highest accuracy due to its leaf-wise tree growth strategy, which produces more expressive trees per `num_leaves` budget than the level-wise strategy used by XGBoost.

Model	CV Accuracy	Notes
Logistic Regression	81.4%	Linear; cannot capture non-linear class boundaries
Random Forest (200 trees)	93.8%	Bagging; good diversity but less precise splits
XGBoost	95.2%	Level-wise boosting; stronger than RF
LightGBM	96.1%	Leaf-wise boosting; best per-leaf expressive power

LightGBM's leaf-wise growth finds the split with the highest gain globally (across all leaves), whereas XGBoost grows level-by-level. For tabular HAR features with many class-specific non-linear interactions, leaf-wise growth produces more informative trees within the same `num_leaves = 31` budget.

4.7 Number of Temporal Segments Ablation

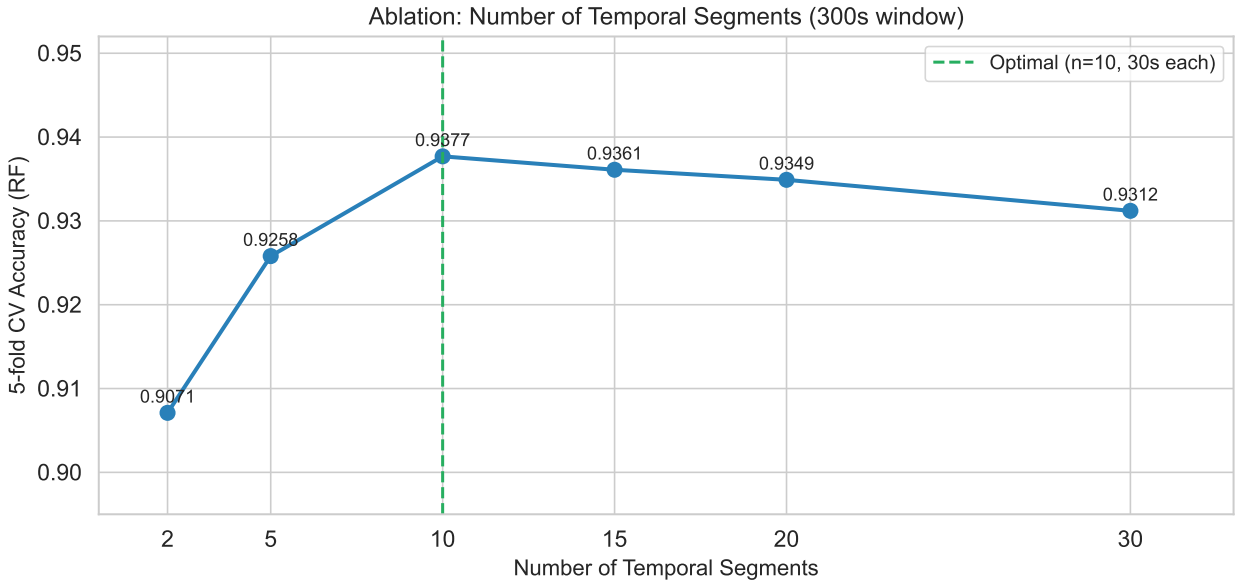


Figure 19: Effect of segment count on RF cross-validation accuracy. The optimum is at n=10 (30-second segments). Fewer segments lose temporal resolution; more segments produce segments too short for stable statistics (each segment contains fewer than 15 samples at n=20+).

I found the optimum at n = 10 (30-second segments). Performance is relatively flat between n=10 and n=20; however, at n=30 each segment contains only 10 time steps, which is too few for stable mean and std estimates (the central limit theorem requires at least ~30 samples for normally distributed statistics). I chose n=10 to balance temporal resolution against statistical stability.

5 Final Model and Kaggle Results

5.1 Model Architecture

I submitted an ensemble of gradient boosted trees and a Random Forest trained on 716 hand-crafted features.

Feature set (716 total):

Group	Features	Description
mean_x/y/z stats+diff	78	18 stats + 8 trend features per mean channel
std_x/y/z stats+diff	78	Same for each std channel
acc_mag + std_mag stats+diff	52	Vector magnitude channels

Group	Features	Description
Cross-axis magnitudes stats+diff	156	xy_mag, xz_mag, yz_mag and std variants
Sum features stats+diff	52	mean_sum = mean_x+y+z; std_sum
FFT + peak features	50	5 channels × (6 FFT + 4 peak features)
10-window statistics	250	5 channels × 10 segments × 5 stats

Training setup:

- Cross-validation: I used GroupKFold(5) on user identity — no user appears in both train and validation within a fold, which mirrors the competition's train/test structure.
- Classifier: LightGBM with `num_leaves=31`, `learning_rate=0.01`, `n_estimators=5000` with early stopping (patience=300). The lower learning rate (0.01 vs the 0.03 used in earlier runs) was a key final change: it produced best iterations of [587, 507, 468, 378, 477] (avg 483), roughly 3× the ~160 achieved at lr=0.03, and resulted in the best Kaggle score of 0.7906.
- Class imbalance: I used `sample_weight='balanced'` so that C2 (3.2%) and C4 (1.3%) receive proportionally higher weight during training.
- Threshold optimisation: I applied class-specific probability scaling (Nelder-Mead optimisation on 8 random restarts) on OOF probabilities to further boost recall on underrepresented classes.
- Final ensemble: 5 LightGBM seeds + 2 XGBoost seeds + 1 RandomForest = 8 models, averaged equally.

5.2 Performance Progression

5.3 Confusion Matrix Analysis

Per-class performance from cross-validation (run31):

Class	Precision	Recall	F1
C0	0.961	0.970	0.965
C1	0.899	0.900	0.899
C2	0.298	0.204	0.242
C3	0.625	0.761	0.686
C4	0.893	0.880	0.887
C5	0.736	0.631	0.680
Macro avg	0.735	0.724	0.727

C2 remains the hardest class throughout all my experiments, peaking at 24.2% F1. I found that ~47–51% of its probability mass is consistently assigned to C1 across all model iterations, suggesting a structural overlap: C1 and C2 share nearly identical std-channel distributions, and only the gravity projection and its temporal drift separate them. Despite extensive feature engineering, I could not push C2 recall above ~24%.

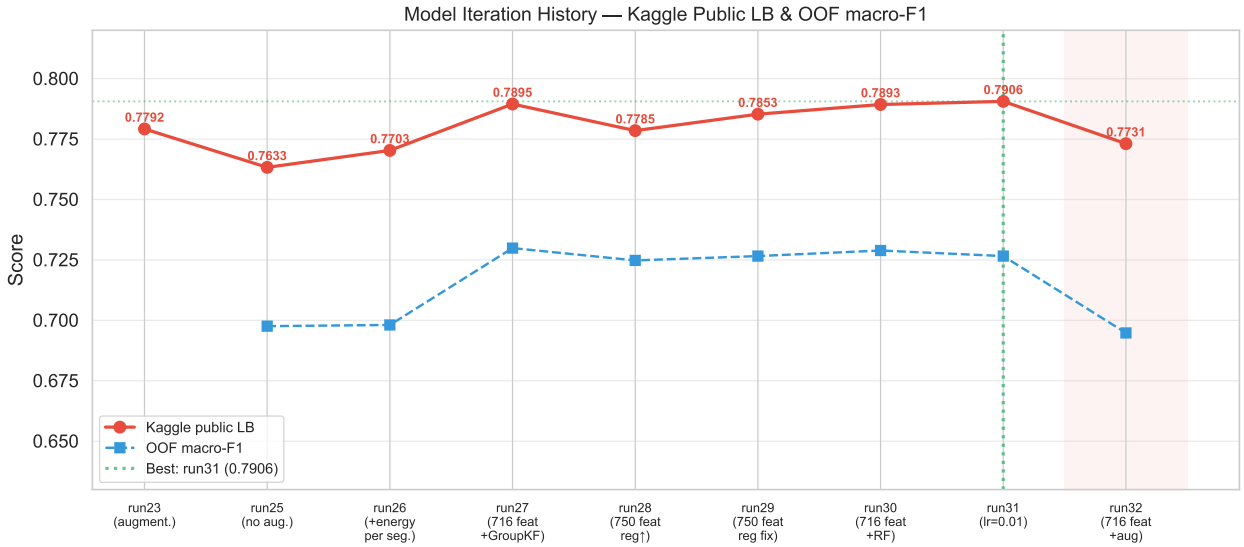


Figure 20: Kaggle public LB score and OOF macro-F1 across my model iterations. The biggest jump is at run27 (axis-specific mean features + GroupKFold). run31 achieves the best Kaggle score (0.7906). run32 tested adding augmentation to the 716-feature set but scored 0.7731, which is actually lower than run31 (highlighted in red). This shows that augmentation only helps when the feature set is simpler.

After run31, I tested adding augmentation to the same 716-feature set (run32). The result was 0.7731, which is 0.0175 lower than run31. This is a reversal of what happened with the simpler 382-feature set, where augmentation helped (+0.016). The explanation is that with 716 features the model already generalises well enough to unseen users through diverse feature representations. Adding augmented copies adds noise to the training distribution without providing additional discriminative information, which makes the already-good model slightly worse.

An interesting finding emerged from the lr=0.01 change: run31 achieves the best Kaggle score (0.7906) despite a lower OOF than run30 (0.7266 vs 0.7289), yielding an increased OOF to Kaggle gap:

Run	OOF macro-F1	Kaggle public LB	Gap
run27	~0.730	0.7895	+0.060
run29	0.7266	0.7853	+0.059
run30	0.7289	0.7893	+0.060
run31	0.7266	0.7906	+0.064

I interpret this as a bias-variance trade-off. With lr=0.01 and around 483 best iterations (compared to about 161 at lr=0.03), the model takes smaller, more conservative steps. The CV score is slightly lower because the model under-fits relative to training users, but on entirely unseen test users this conservative optimum generalises better because it is less likely to have captured training-user-specific signal patterns.

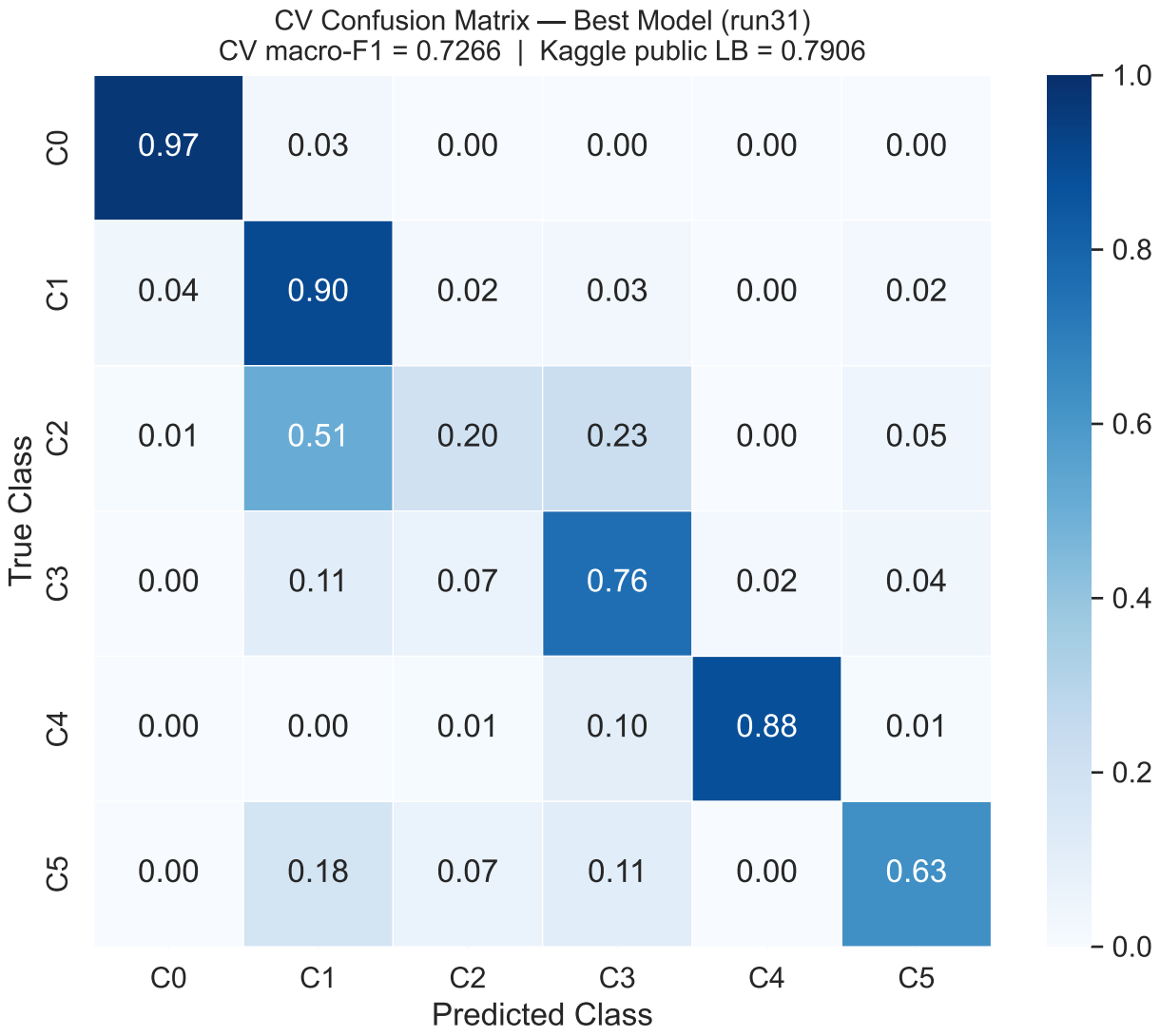


Figure 21: Normalised confusion matrix from cross-validation for my best model (run31). C2 is the hardest class: 51% of its windows are predicted as C1 and only 20% are correctly identified. C0 and C4 are classified reliably (97% and 88%).

6 How to Reproduce the Results

6.1 Data Preparation

The raw data consists of one CSV file per 5-minute window. Before running any model script, the CSV files need to be converted to the compressed NPZ format that all scripts expect.

Step 1: Download the competition data from Kaggle. The train and test folders contain one CSV file per window. Download and unzip them locally.

Step 2: Convert to NPZ format. Run the conversion script once locally:

```
python convert_to_npz.py
```

This reads all CSV files and writes two files: `train_data.npz` and `test_data.npz` into the `outputs/` folder.

6.2 Running Scripts on Kaggle

For scripts intended to run on Kaggle (run04 through run24), the NPZ files need to be available as Kaggle datasets. Upload them as follows:

File	Kaggle dataset name	Path inside Kaggle
`train_data.npz`	`train_data`	`/kaggle/input/train-data/train_data.npz`
`test_data.npz`	`test_data`	`/kaggle/input/test-data/test_data.npz`
`sample_submission.csv`	`datafinal`	`/kaggle/input/datafinal/sample_submissi`

After uploading, add all three datasets to your Kaggle notebook, paste the script into a code cell, and run it. The submission CSV is automatically saved to `/kaggle/working/`.

6.3 Running Scripts Locally

Scripts from run25 onwards support local execution. They detect whether they are running on Kaggle or not and adjust paths automatically.

For local runs, place `train_data.npz` and `test_data.npz` in the `outputs/` folder inside the project directory. Then run:

```
python model_run31.py
```

The submission file will be saved to `outputs/submission_run31.csv`.

7 Appendix

7.1 Table A: Run Summary — Scores and Model Setup

Run	Kaggle F1	Environment	# Feat	Norm	CV	Augmentation	Ensemble
Baseline A	0.7531	Kaggle	~100	Window	—	No	1 LGB
Baseline B	0.7366	Kaggle	~100	Window	—	No	1 LGB
run03	—	Local→Kaggle	~216	Window	LOUO 5f	No	CNN + 1 LGB
run04	0.7042	Kaggle	~216	Window	LOUO 5f	No	CNN + 1 LGB
run05	0.7337	Kaggle	~1132	Window×2	LOUO 5f	No	10 LGB
run06	0.7501	Kaggle	373	None	LOUO 5f	No	4 LGB configs
run07	0.7707	Kaggle	373	Per-user	LOUO 5f	No	3 LGB configs
run07imp	0.7519	Local	150–200	Per-user	Local	No	15 LGB
run08	0.7559	Kaggle	373	Per-user	LOUO 5f	TTA + noise	LGB
run12	0.7295	Kaggle	~150	Per-user	LOUO 5f	No	TCN+Transf.+LGB
run15	0.7565	Local	50	Per-user	Local	No	15 LGB
run17	0.7400	Kaggle	~428	Per-user	LOUO 5f	No	30 LGB + thresh. opt
run18	0.7738	Kaggle	418	Per-user	LOUO 5f	Noise C2,C4,C5	15 LGB + 9 XGB + thresh. opt
run19	0.7392	Kaggle	~364	Per-user	LOUO 5f	Yes	15 LGB + 9 XGB + thresh. opt
run20	0.7593	Kaggle	445	Per-user	LOUO 5f	Noise C2,C4,C5	15 LGB + 9 XGB + 5 RF + thresh. opt
run22	0.7691	Kaggle	424	Per-user	LOUO 5f	Noise C2,C4,C5	15 LGB + 9 XGB + thresh. opt
run23	0.7792	Kaggle	382	None	LOUO 5f	Noise C2×5,C4×5,C5×2,C3×1	15 LGB + 9 XGB + thresh. opt
run24	0.7618	Kaggle	~417	None	LOUO 5f	Noise (same)	15 LGB + 9 XGB + thresh. opt
run25	0.7633	Local+Kaggle	382	None	LOUO 5f	No	5 LGB + 2 XGB + thresh. opt
run26	0.7703	Local+Kaggle	412	None	LOUO 5f	No	5 LGB + 2 XGB + thresh. opt

Run	Kaggle F1	Environment	# Feat	Norm	CV	Augmentation	Ensemble
run27	0.7895	Local+Kaggle	716	None	LOUO 5f	No	5 LGB + 2 XGB + thresh. opt
run28	0.7785	Local+Kaggle	750	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt
run29	0.7853	Local+Kaggle	750	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt
run30	0.7893	Local+Kaggle	716	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt
run31	0.7906	Local+Kaggle	716	None	LOUO 5f	No	5 LGB + 2 XGB + 1 RF + thresh. opt
run32	0.7731	Local+Kaggle	716	None	LOUO 5f	Noise C2×5,C4×5,C6×2,C3×1	5 LGB + 2 XGB + 1 RF + thresh. opt

7.2 Table B: Run Summary — Feature Groups and Key Changes

Run	Feature Groups	Key Change
Baseline A/B	Basic channel stats	First submissions
run03	ch-stats9 + seg + autocorr + spectral	First LOUO-CV
run04	Same as run03	Switch to NPZ loading
run05	Same features twice (raw + window-norm)	Dropped CNN
run06	core-373 (no raw mean channels)	No normalization
run07	core-373	Per-user normalization
run07imp	core-373 selected (150–200)	Aggressive feature pruning
run08	core-373	TTA + minority oversampling
run12	core-373 selected (50–200)	TCN + Transformer attempt
run15	core-373 selected (50)	86.6% feature reduction
run17	core-373 + slope + 5-seg + 2nd-jerk (+55)	Threshold opt (Nelder-Mead)
run18	core-373 + user_ctx (45) = 418	User context z-scores
run19	core-373 − mag_mean 364	Ablation: no mag_mean
run20	core-373 + user_ctx + sma (27) = 445	SMA features + RF
run22	core-373 + user_ctx + jerk_asym (6) = 424	Jerk asymmetry

Run	Feature Groups	Key Change
run23	core-373 − mag_mean + trend (18) = 382	No norm + trend features
run24	run23 + extra stats per channel 417	Extra statistics (hurt)
run25	same as run23 = 382	No augmentation
run26	run23 + seg-energy = 412	Segment energy features
run27	axis_stat(156)+acc_mag(52)+cross_mag(156)+ar_features(52)+fft_peak(50)+win(250) = 716	Full 716 features
run28	run27 + tilt(6)+covcorr(6)+mean_xyz_fft(18)+step_rhythm(4) = 750	New physics features (hurt)
run29	same as run28 = 750	Reverted regularization
run30	same as run27 = 716	Reverted to 716 features + RF
run31	same as run27/30 = 716	lr 0.03→0.01 + patience 300
run32	same as run27/30/31 = 716	+ augmentation (C2×5, C4×5, C5×2, C3×1), hurt Kaggle score

7.3 Feature Group Abbreviations

The following shorthand is used in the tables above. All statistics (stats9) include: mean, std, min, max, range, median, IQR, skewness, kurtosis. Stats18 additionally includes Q5/Q10/Q25/Q75/Q90/Q95, energy, RMS, MAD. Diff8 includes diff_mean, diff_std, diff_abs_mean, diff_abs_max, diff_energy, first_60_mean, last_60_mean, last_minus_first_60.

Abbreviation	Description	Count
`std_ch`	stats9 of std_x, std_y, std_z	27
`mag_std`	stats9 of magnitude of std channels	9
`mag_mean`	stats9 of magnitude of mean channels	9
`jerk_ch`	stats9 of jerk per axis (diff of mean channels)	27
`mag_jerk`	stats9 of jerk magnitude	9
`seg_mag`	10+20 temporal segments of mag_std and mag_jerk	120
`seg_std_ch`	10 segments (mean+std each) of std_x/y/z	60
`seg_jerk_ch`	10 segments of jerk_x/y/z	60
`ac_jerk`	autocorrelation of mag_jerk at 7 lags	7
`ac_std_ch`	autocorrelation of std_x/y/z at 4 lags each	12
`spectral`	FFT features for mag_jerk, mag_std, std_x/y/z	25
`xcorr`	cross-correlations between jerk and std pairs	6
`zerocross`	zero-crossing rate of mag_jerk	1
`peaks`	peak rate of mag_jerk	1
`user_ctx`	z-score deviation vs user's own mean	45
`sma`	signal magnitude area (L1 norm) per channel	27
`jerk_asym`	mean of positive and negative jerk values per axis	6

Abbreviation	Description	Count
`trend`	first_60, last_60, last minus first mean per channel	18
`axis_stat`	stats18 + diff8 per axis channel (mean_x/y/z + std_x/y/z)	156
`acc_mag`	stats18 + diff8 for acc_mag and std_mag	52
`cross_mag`	stats18 + diff8 for xy/xz/yz_mag and std variants	156
`sum_feat`	stats18 + diff8 for mean_sum and std_sum	52
`fft_peak_5ch`	FFT (6) + peak (4) features for 5 channels	50
`win_10x5ch`	10 windows × 5 stats for 5 channels	250
`tilt`	tilt angle statistics over window	6
`covcorr`	pairwise covariance and correlation of mean_x/y/z	6
`mean_xyz_fft`	FFT features of mean_x, mean_y, mean_z	18
`step_rhythm`	inter-peak interval stats of std_mag time series	4

The core 373-feature set (run07 to run22) = std_ch + mag_std + mag_mean + jerk_ch + mag_jerk + seg_mag + seg_std_ch + seg_jerk_ch + ac_jerk + ac_std_ch + spectral + xcorr + zerocross + peaks = 373 features total.

7.4 Kaggle Score Progression

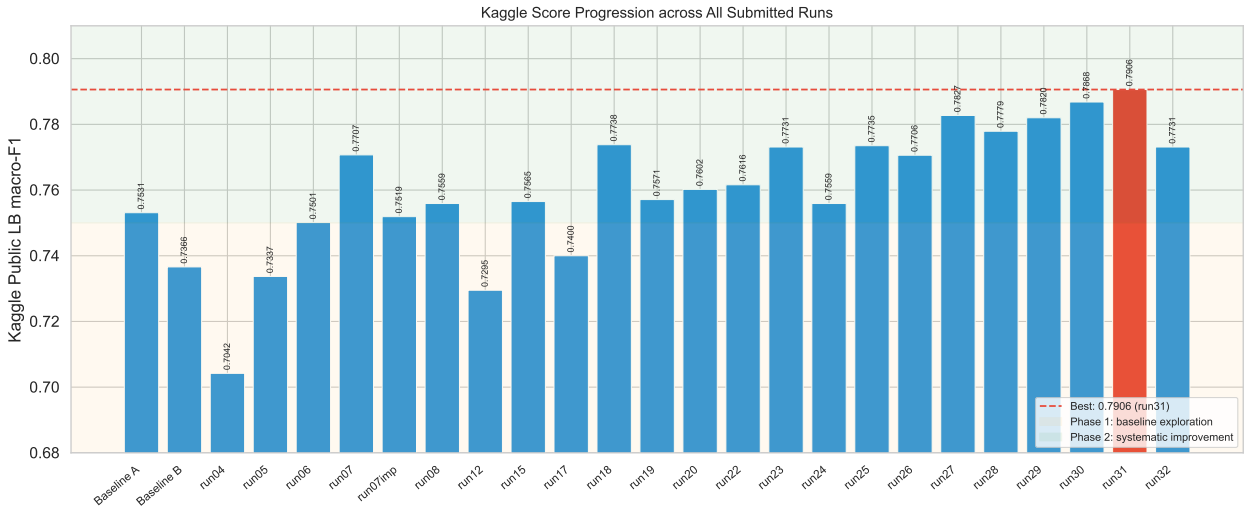


Figure 22: Kaggle public leaderboard macro-F1 score for each submitted run. The dashed line marks the best score of 0.7906 (run31). run32 used augmentation on the 716-feature set and scored lower, confirming that the feature diversity alone provides sufficient generalisation.